

# Assignment Description

## Equipment Loan Management System for an IT Department

Your task is to design and implement a simple web application for managing equipment loans within an IT department. The system should allow users to register and track the lending of equipment such as laptops, monitors, headsets, keyboards, mobile phones, and other technical devices.

You should plan and implement as much functionality as is realistically achievable within the available time. Mandatory requirements should be prioritized first.

The solution should focus on simplicity, ease of administration, and providing a clear overview of equipment. There is no requirement to implement advanced role hierarchies, integrations with external systems, or complex business logic beyond what is necessary to satisfy the requirements.

### Technical Requirements

- The application must be a web application accessible through modern web browsers.
- The application must provide persistent data storage (e.g., using a database).
- The application must support user authentication (login).
- The application must implement authorization so that only authenticated users can access the system.
- You are free to choose the programming language, framework, and libraries, but your choices must be documented and justified.
- The application should provide a user-friendly interface optimized for desktop use.
- Source code must be committed throughout development to a version control system such as GitHub.

### Functional Requirements

- The application must be a web application accessible through modern web browsers.
- The application must provide persistent data storage (e.g., using a database).
- The application must support user authentication (login).
- The application must implement authorization so that only authenticated users can access the system.
- You are free to choose the programming language, framework, and libraries, but your choices must be documented and justified.
- The application should provide a user-friendly interface optimized for desktop use.
- Source code must be committed throughout development to a version control system such as GitHub.

### **[R-1] Equipment Overview**

Users must be able to view a list of all equipment registered in the system.

Each equipment item should display:

- Name
- Category
- Serial number or asset number (if available)
- Status
- Availability (Available or Checked Out)
- Example statuses:

- Available
  - Checked Out
  - Under Maintenance
  - Users should be able to filter the list by:
  - Name (free-text search)
  - Category
  - Status
  - Borrower (for checked-out equipment)
  - Users should also be able to sort the list by one or more of the following:
  - Name
  - Category
  - Status
  - Users must be able to select an equipment item to view its details (**R-2**).
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### **[R-2] Equipment Details**

Users must be able to view detailed information about a selected equipment item.

The detail page should include:

- Name
  - Category
  - Serial number or asset number
  - Description
  - Status
  - Current borrower (if checked out)
  - Check-out date (if checked out)
  - Due date (if applicable)
  - Loan and return history
  - Each history entry should display:
  - Event type (Check-out or Return)
  - Date and time
  - User who registered the action
  - Borrower
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### **[R-3] Create, Edit, and Delete Equipment**

Users must be able to create, edit, and delete equipment records.

When creating or editing equipment, the following fields should be available:

- Name (required)
- Category (required)
- Serial number or asset number (*optional but recommended*)
- Description (*optional*)
- Status (required)

Deletion must be implemented as a soft delete, meaning equipment is marked as deleted instead of being permanently removed from the database. This preserves historical records and allows previous transactions to be tracked.

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### **[R-4] Register Equipment Check-Outs and Returns**

Users must be able to register equipment check-outs and returns.

When registering a check-out, the following information should be stored:

- Equipment being checked out
- Borrower
- Check-out date and time
- Due date (*optional but recommended*)
- User who recorded the transaction

When registering a return, the following information should be stored:

- Equipment being returned
- Return date and time
- User who recorded the transaction

After registration:

- Equipment status should be updated automatically.
  - The active loan should be marked as completed upon return.
  - The loan history should be updated and displayed in the equipment details.
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#### **[R-5] Authentication and Authorization**

The application must include user authentication and access control.

Requirements:

- Unauthenticated users should only have access to the login page.
- Users must be able to log in using a username or email address and password.
- Authenticated users should have access to the application's functionality.
- User registration is not required.
- Advanced role management is not required, although candidates may optionally distinguish between employees and administrators if implemented in a simple and appropriate manner.

#### Scope and Limitations

The following features are not required:

- Mobile responsiveness
  - Integration with Active Directory, Microsoft Entra ID, email, or other external systems
  - Barcode scanning
  - Image or document uploads
  - Advanced role and permission management
  - Automatic notifications or reminders
  - Full accessibility compliance beyond a clean and understandable user interface
  - Advanced error handling beyond standard input validation and user-friendly error messages
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This was a practical assignment for my Vocational Qualification Examination (Fagprøve). The objective was to design and build a functional equipment loan management system while demonstrating competence in full-stack web development, authentication, database design, and software engineering best practices within a limited time frame.